

Lesions Involving Medial Anterior Forebrain Pathway Circuitry Destabilize Phrase Timing in Adult Canary Song

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Abstract

Basal ganglia–thalamo-cortical circuits are essential for learning complex motor sequences, yet their roles in controlling flexible motor behavior remain poorly understood. The homologous songbird Anterior Forebrain Pathway (AFP) drives song motor learning, and was previously thought not to play a role in song performance, as early lesion studies reported no detectable effect. This perspective is now debated, as newer results demonstrate effects in subsets of songbirds. Here, we revisit this question in adult canaries by performing bilateral excitotoxic lesions targeting the lateral and medial subdivisions of the AFP.

To quantify behavioral changes across thousands of recorded canary songs, we developed a high-throughput annotation pipeline that extends a self-supervised vision transformer (TweetyBERT) with a supervised classification head, eliminating the memory bottleneck of UMAP-based clustering. This model enables phrase-level analysis across thousands of songs per bird. We find that lesions involving the medial AFP produce a stuttering-like behavior, characterized by a significant increase in the variability of phrase durations. This effect is absent in birds with only lateral AFP lesions. The increase in variability was sustained across the post-lesion recording period and was accompanied by detectable changes in syllable acoustic structure.

Our results implicate the medial AFP in the ongoing control of phrase duration in adult canary song, challenging the view that the AFP is dispensable once song is learned. These findings position the medial AFP as a tractable model for understanding how basal ganglia and cortical dynamics jointly maintain complex learned motor sequences.

Keywords

Basal ganglia, Anterior Forebrain Pathway, stuttering, systems neuroscience, sensorimotor, motor sequences, flexible motor behavior

Introduction

Complex motor skills, such as speaking, require the ability to combine simpler motor elements into flexible sequences. Saying the word “cat” requires the speaker to plan the sequence of phonemes in the correct order and coordinate their vocal organs to produce each sound in the sequence. How these sequences are planned and executed remains an open question in neuroscience (Lashley, 1951). Damage or deterioration of the brain regions and pathways responsible for motor planning is implicated in debilitating motor disorders, such as Parkinson's Disease and Huntington's Disease, as well as speech disorders like stuttering (Craig-McQuaide et al., 2014).

Birdsong serves as a model for investigating how these brain regions plan and execute sequences of actions. Songbird neural circuitry is more simply connected than mammalian systems, yet remains analogous to mammalian motor control regions and cell types (Jarvis, 2019; Reiner et al., 2004). Canary song is composed of syllables that the bird repeats in a “phrase” for about one second, before switching to the next syllable in a sequence. These sequences have long-range order; what the bird sings at one step influences what it may sing up to five to seven phrases later (Markowitz et al., 2013). Thus, the varying complexities across songbirds enable neuroscientists to investigate how brain regions orchestrate motor sequences of varying complexity and at different scales: syllable, phrase, and song (Fig. 2B).

Birdsong is also a fundamental model for studying the reinforcement learning of complex motor sequences. Learning is driven by nuclei in the lateral portion of the Anterior Forebrain Pathway (AFP) (Fig. 1A); the cortical nucleus LMAN drives variability, allowing for motor exploration, and the basal ganglia nucleus Area X directs song learning towards improved performance (Fee & Goldberg, 2011). Research has focused primarily on the lateral AFP, but histological evidence suggests a parallel medial AFP exists, comprised of the cortical nucleus MMAN (magnocellular nucleus of the anterior nidopallium), the medial portion of Area X, and the thalamic nucleus DLM (Hara et al., 2007; Nottebohm et al., 1976). The potential role of medial AFP in adult song, particularly in timing and sequencing, is a central question of this study.

Early studies suggested the AFP was dispensable in adult song, as lesions produced no clear impact in zebra finches (Bottjer et al., 1984; Sohrabji et al., 1990) nor canaries (Nottebohm et al., 1976). More recent work has challenged this view. Lesions to the lateral portion of Area X produced stuttering in a subset of adult zebra finches that repeated syllables at the end of their songs pre-lesion (Kubikova et al., 2014), while partial Area X lesions in Bengalese finches produced transient stuttering-like behavior that resolved within 21 days and showed no clear correlation with lesion size (Kobayashi et al., 2001). Lesions to MMAN in Bengalese finches resulted in greater variability in syllable repeat number, and increased uncertainty in the order of syllables (Koparkar et al., 2024). Together, these results suggest that the AFP continues to influence adult birdsong.

Our experiments expand upon previous findings by performing bilateral lesions to both the lateral and medial portions of the AFP in adult canaries during their breeding season. To quantify behavioral changes across the large number of songs produced, we developed a high-throughput annotation pipeline built on TweetyBERT, a self-supervised model that clusters

canary syllables with high accuracy (Vengrovski et al., 2026), overcoming the scalability limitations of UMAP to enable analysis of thousands of songs per bird. Medial and lateral AFP lesions increased phrase duration variability in select syllables, and produced minor changes in the acoustic structure of those syllables. These effects were absent in lesions targeting only the lateral AFP and negative controls. These findings suggest that the medial AFP continues to influence the timing of transitions between repeated syllables in adult canary song, positioning the medial thalamocortico-basal ganglia pathway as a valuable model for understanding basal ganglia and cortical dynamics in motor sequencing.

Materials and Methods

I. Experimental Design

Subjects, Housing, and Husbandry.

Subjects were adult male canaries ($n = 17$ lesion, $n = 4$ sham negative control, age > 1 year) obtained from Maryland Exotic Birds (Pasadena, MD). While in holding, birds were housed in flight cages on a light cycle synchronized to daylight hours in Eugene, Oregon. Birds were provided with food (Kaytee Supreme Seed, Lafeber's Canary Premium Daily Pellets), fresh water, and high-calcium grit (Kaytee Hi-Calcium Grit Supplement) *ad libitum*. Animal housing and husbandry were performed by the University of Oregon's Terrestrial Animal Care Services (TeACS). All care and experimental manipulation protocols were reviewed and approved by the University of Oregon Institutional Animal Care and Use Committee (IACUC).

Experimental Animal Housing and Acoustic Recording

For experimental recordings, birds were singly housed in double-breeding cages with the divider removed to provide additional space (Model C-1510, Kings Cages International, Ft. Lauderdale, FL). Cages were placed inside Environmental Chambers (Omnitech Electronics Inc., Columbus, OH), which were lined with soundproofing foam to reduce external noises (Willtec Melamine SONEX Classic Noise Absorbing Foam Panels, Sonex, Minneapolis, MN). Fresh air was pumped into the chambers using tubes connected to a small air pump (Model #LPH80, JEMCO). Light cycle timing was adjusted to match the corresponding daylight hours in Eugene, Oregon. Enrichment for experimental birds included socialization (1-2 hours per day, 3-5 times per week) and daily treats and toys (dried mealworms, dried fruit, crinkle paper, and mixed greens).

Song recordings were collected using an omnidirectional microphone (Audio-Technica AT803) positioned above each cage. An M-Audio 8 pre-amplifier was used to record from each channel and route recordings to the computer. Songs were detected, then recorded using Sound Analysis Pro 11 software (Tchernichovski et al., 2000). Recordings were conducted at 44.1 kHz (single channel) and saved as WAV files. All recorded songs were undirected (sung in the absence of females and other male birds).

Song Detection and Annotation

Song segments were isolated from background noise using a supervised song detector trained on ground-truth labels marking the start and stop of songs, to eliminate calls, cage noise, and

extended silences from raw recordings (Vengrovski et al., 2026). Songs were then annotated using a previously published, self-supervised song annotator that uses a masked-prediction vision transformer model applied to spectrograms (Vengrovski et al., 2026). TweetyBERT embedding space is structurally organized with clear clusters for each canary song syllable, dimensionally reduced using Uniform Manifold Approximation and Projection (UMAP), then Hierarchical Density-Based Spatial Clustering with Noise (HDBSCAN) was used to cluster the UMAP embedding space (Vengrovski et al., 2026).

Because UMAP is both computationally and memory-intensive, the original TweetyBERT annotation software has an upper bound of annotating approximately 45 minutes of birdsong. 45 minutes of audio typically contains every unique instance of each syllable in a canary's repertoire, making it sufficient for initial label discovery; however, it is insufficient for studying the long-term changes in singing required by this study. To extend the power of automated song labeling to larger datasets, we trained a supervised deep neural network to perform syllable classification using pseudolabels generated by TweetyBERT. Using TweetyBERT, we passed and dimensionally reduced approximately 45 minutes of canary song recordings, from which we generated an initial set of canary syllable labels. We then trained a supervised MLP multi-class classification layer on top of the TweetyBERT network to fit the initial set of discovered clusters. After training, the model can process spectrograms and reproduce the discovered UMAP + HDBSCAN states without the slow, memory-intensive UMAP step, enabling it to classify datasets of arbitrary size.

We applied simple post-processing techniques to improve annotation quality and reduce noisy labels. We merged detected songs that were less than 500 ms apart into a single song for analysis. If the same syllable was on either side of a merge, and less than 10 ms apart, this syllable annotation label was merged. This post-processing was meant to prevent songs that got split up by the song detector from breaking up stretches of repeated syllables. Phrase duration was defined as the amount of continuous time bins marked with a given syllable label (Fig. 2B).

Lesion and sham lesion surgeries

Surgical protocols were based on previously published lesion protocols targeting song nuclei in the Anterior Forebrain Pathway (AFP) (Goldberg & Fee, 2011; Koparkar et al., 2024). Birds were induced with 3-5% isoflurane (Isospire, Dechra Veterinary), anesthetized with Ketamine (Zetamine ketamine hydrochloride injection, VetOne) and Midazolam injection (USP), and maintained at 1-3% isoflurane for the remainder of the surgery. Birds were mounted in a stereotaxic device (Kopf), stabilized by non-rupture ear bars and a custom beak bar, and positioned so that the head angle of the anterior, flat portion of the skull was set to 20 degrees relative to the surface of the surgery table. A small craniotomy was made (diameter less than 0.5 mm), followed by a small durotomy (diameter less than 0.1 mm) over the injection site. Bilateral excitotoxic lesions were made by injecting N-methyl-D-aspartic acid to target Area X (NMDA; Sigma, St. Louis, MO). Our injection coordinates were based on previous lesions targeting Area X in zebra finches (Goldberg & Fee, 2011). Volumes, injection number, and injection location were then adjusted using histology to better target Area X in canaries. The outer skull leaflet above the midsagittal sinus ("lambda") was removed, and stereotaxic

coordinates were set to zero at this point. Injections were typically between 5.40-5.60 mm anterior to lambda, 0.85-1.60 mm lateral to the skull's midline, and 2.60-3.70 mm ventral to the surface of the brain. Sham lesions were made by injecting sterile saline solution (Hospira 0.9% sodium chloride injection, USP).

Histology and lesion assessment

Birds were anesthetized using isoflurane, then sacrificed using an overdose of pentobarbital (Euthasol, 150 mg/kg), shortly before a transcardial perfusion with 1x Phosphate Buffered Saline (Growcells.com MRGF-6396, diluted) for 4 minutes at a 7.56 flow rate, followed by fixation in 4% PFA diluted in 1x PBS for 10 minutes at a 7.56 flow rate (Rabbit-Plus Peristaltic Pump, Rainin). Brains were stored in 4% PFA overnight, then transferred to fresh 4% PFA for 24-48 hours, then to 1x PBS for 1-7 days before sectioning.

Brain tissues were embedded in 2.5% agarose (Sigma-Aldrich, AG6013-25G Type I, low EEO Agarose) mixed with 1x PBS. Brains were sectioned using a Leica VT 1200S vibratome in either 50 μ m or 75 μ m thick sections, mounted onto slides (Fisherbrand Superfrost Plus Microscope Slides), and left to dry overnight. Slides were then rehydrated in a graded ethanol series, stained with diluted thionin solution, dehydrated in a graded ethanol series, cleared with HistoClear, and coverslipped with mounting media (Eprelia Shandon-Mount). This protocol was based on other staining protocols for visualizing AFP song nuclei in seasonal songbirds (Singh et al., 2003; Smith et al., 1997), as well as a publicly available Nissl staining protocol (Huang, n.d.). Slides were imaged at 5x magnification on a Leica Widefield Scope, and the mosaic was stitched using LASX and Thunder software. The Nissl-defined borders of song nuclei and the striatal lamina were identified, and lateral or medial Area X hit types were observed and noted qualitatively (Fig. 1).

II. Statistical Analysis

Quantifying changes to phrase duration variability

We tested whether the stuttering-like phenotype differed across lesion hit types: medial and lateral AFP hits, lateral AFP-only hits, and sham saline negative controls. Phrase duration was defined as the continuous interval during which a syllable label persisted before transition to the next syllable. In a representative bird, post-lesion song showed prolonged repetition of a single syllable that was not evident pre-lesion (Figure 2B,C). Because the lesion effect was restricted to a subset of syllables rather than distributed uniformly across the repertoire (Figure 2C), we focused variability analyses on the syllables most likely to express the phenotype. For each bird, we selected the top 30% highest-variability syllables, ranked by post-lesion phrase duration variability, to capture syllable-specific disruption without averaging it away across largely unaffected syllables.

To quantify variability in phrase duration, we calculated the standard deviation of phrase durations for each syllable within each bird for early pre-lesion, late pre-lesion, and post-lesion epochs. To prevent differences in sample size from biasing estimates of variability, phrase duration measurements were usage-balanced within each bird $\times$ syllable comparison. For each

syllable, we randomly subsampled the late pre-lesion and post-lesion phrase duration distributions to an equal number of renditions before calculating variance and standard deviation. Variability was then quantified as the standard deviation of phrase durations, allowing pre- and post-lesion variability to be compared in milliseconds. In aggregate pre-lesion versus post-lesion comparisons, lesion groups showed greater dispersion and more observations above the identity line than sham controls, consistent with increased post-treatment phrase duration variation (Fig. 3A).

For lesion hit type comparison, the standard deviations for the top 30% highest variability syllables were averaged within each animal, so that each bird contributed one value per lesion hit type. The animal-level aggregation was used to avoid treating multiple syllables from the same bird as independent biological replicates. Complete medial and lateral lesions and partial medial and lateral lesions were pooled into a single medial + lateral lesion group for the primary comparison because both lesion types involved damage to medial AFP circuitry. The remaining groups were lateral AFP only lesions and sham saline negative controls. Within each lesion-hit group, we tested whether phrase duration standard variation increased from late pre-lesion to post-lesion using paired tests comparing each group's standard deviations. We tested whether the change in standard deviation differed between each lesion group and sham controls using one-tailed Welch's t-tests on the change metric, with the alternative hypothesis that lesion groups would show a greater post-treatment increase in phrase duration variability than sham controls. Significance annotations in the figure denote these planned comparisons. This analysis tested whether lesions involving medial AFP produced a larger increase in phrase duration variability than sham treatment or lesions limited to lateral AFP.

Phrase duration variability was quantified for each animal and syllable by comparing the standard deviation of phrase durations between the late pre-lesion and post-lesion epochs. To focus on syllables most likely to express lesion-associated timing instability while reducing reliance on post-lesion-only selection, syllables were selected within each animal using a pooled variability criterion. Specifically, for each animal $\times$ syllable pair, we calculated the mean of late pre-lesion and post-lesion phrase duration variance and retained the top 30% of syllables per animal. Phrase duration standard deviation was then expressed in seconds for visualization and statistical analysis.

For scatterplots, each point represents one selected animal $\times$ syllable pair. For group-level statistical tests, selected syllables were first averaged within each animal so that each bird contributed one independent value per epoch. Late pre-lesion and post-lesion phrase duration SDs were compared within each lesion group using one-tailed paired t-tests testing the directional hypothesis that post-lesion variability increased. Sham, lateral-only, and medial+lateral lesion groups were analyzed separately, with partial and complete medial+lateral lesions pooled for the primary behavioral summary.

To visualize the time course of the lesion effect, daily phrase duration variance values were converted to standard deviation and expressed in seconds. The daily time-course table was filtered to the same animal $\times$ syllable pairs selected for the late pre-lesion versus post-lesion analysis. Within each lesion group, daily values were summarized using a 7-day rolling median

and interquartile range. The smoothing window was applied separately before and after lesion day so that pre-lesion and post-lesion values were not averaged together.

To assess whether the phrase-duration variability result depended on the high-variability syllable selection criterion, we repeated the analysis using the top 20%, 30%, and 40% most variable syllables per bird. To address potential circularity from selecting syllables based on post-lesion variability, we also performed an independent-selection validation in which syllables were selected using pre-lesion variability alone and then tested for post-lesion changes in phrase-duration standard deviation. All robustness analyses used the same animal-level statistical framework as the main analysis.

To investigate the time course of the lesion effect, daily phrase duration variance values were converted to standard deviation, and syllable-level values were summarized within each animal. To reduce day-to-day noise and visualize sustained trends, we applied a 7-day rolling median across days within each lesion group. The smoothing window was applied separately before and after lesion day so that pre-lesion values were not averaged together with post-lesion values (Fig. 3E).

Quantifying AFP lesions increases syllable acoustic variability.

To quantify acoustic variability of syllables, we calculated the overlap of syllables between pre-lesion and post-lesion datasets in TweetyBERT latent space using the Bhattacharyya coefficient (Bhattacharyya, 1943). For each comparison, latent embeddings from the two epochs were first subsampled to equal sizes to prevent sample-size differences from biasing the overlap. The two sets of embeddings were then converted into two-dimensional density histograms over the same UMAP bin edges. Each histogram was normalized to sum to one, producing discrete probability distributions p_i and q_i over UMAP space. The Bhattacharyya coefficient (BC) was computed as:

$$BC(p, q) = \sum_i \sqrt{p_i q_i}$$

Where p_i and q_i are the normalized densities of the two distributions in bin i . The Bhattacharyya Coefficient ranges from 0 to 1, with values closer to 1 indicating greater overlap between the two embedding distributions. For the cluster-level summary, BC distributions between pre-lesion and post-lesion were compared within each lesion-hit type using a Wilcoxon signed-rank test (Fig. 4C, D).

Results

Histology

While injections were targeted to the center of Area X, subsequent histology showed that the lesioning area also had spillover into the nidopallium, also lesioning portions of LMAN and possibly MMAN. Thus, our results should be viewed as a statement about the impacts on song from lesioning the Anterior Forebrain Pathway (AFP), rather than on nucleus X specifically. Our

histological results indicate that we hit different regions of Area X and lesioned different volumes. Sham lesions did not affect Area X histology. Some lesions hit only the lateral Area X. Most lesions were partial hits to both the medial and lateral Area X. Some lesions were so large that we could not visualize either the medial or lateral Area X. However, their section location from the midline, and other anatomical features (e.g., proximity to lamina), suggested a complete lesion of both medial and lateral Area X, as well as the visibility of the song nuclei HVC and RA in other Nissl-stained sections (Supplemental Fig.X).

To test whether lesion extent predicted behavioral effect size, we performed an animal-level correlation analysis between effective Area X lesion percentage and the median change in phrase-duration standard deviation. For this analysis, sham birds were assigned 0% effective lesion extent, partial lesions were assigned their measured percent Area X lesioned, and complete medial+lateral AFP lesions were assigned 100%. Each bird contributed one value, computed as the median post-lesion minus late pre-lesion phrase-duration standard deviation across the selected top 30% high-variability syllables. Spearman correlation was used as the primary test.

To test whether lesion location within the AFP influenced post-lesion song behavior, we classified lesions based on their histological relationship to medial and lateral Area X (Figure 1). Nissl-stained sections and anatomical landmarks were used to distinguish medial from lateral Area X and to assign birds to lesion-hit categories (Fig. 1 B,C). Lesions were classified as sham, lateral-only, partial medial and lateral AFP lesions, or complete lesions. These categories were used for subsequent analyses of behavioral and acoustic effects.

Lesions to the medial AFP increased the variation in phrase duration

AFP lesions produced a stuttering-like behavior, characterized by abnormally increased variability in the time spent repeating specific syllables. This effect was not uniform across the repertoire, but was concentrated in a subset of syllables that showed extended post-lesion phrase durations. Therefore, we focused subsequent analyses on the top 30% highest-variability syllables for each bird. This approach was designed to capture syllables that showed the strongest post-lesion disruption while avoiding dilution from the larger number of syllables that were largely unaffected. To test whether phrase duration variability increased after lesion, we compared standard deviations between the late pre-lesion and post-lesion epochs (Fig. 3D). Before calculating variability, syllable usage was balanced across epochs: for each bird-syllable pair, late pre-lesion and post-lesion phrase duration distributions were subsampled to equal sample sizes. This ensured that greater post-lesion variability was not driven by greater numbers of post-lesion song renditions.

Using these usage-balanced measurements, sham saline controls did not show a significant change in phrase duration variability from late pre-lesion to post-lesion, although the median standard deviation increased modestly from 643 ms to 721 ms (paired $t = 0.41$, $p = 0.3544$, $n = 4$ birds). Birds with lateral AFP-only lesions also did not show a significant pre-to-post change, with median standard deviation remaining similar between late pre-lesion and post-lesion epochs (721 ms vs. 708 ms; paired $t = -0.29$, $p = 0.6107$, $n = 8$ birds). In contrast, birds with

complete or partial medial and lateral AFP lesions showed a significant increase in phrase duration standard deviation, with median variability increasing from 672 ms before lesion to 1034 ms after lesion (paired $t = 2.2$, $p = 0.0294$, $n = 9$ birds). These results indicate that increased phrase duration variability occurred when lesions damaged both medial and lateral AFP, whereas lesions limited to lateral AFP did not produce a significant increase in variability.

AFP lesions produced group-specific changes in phrase duration variability. We focused on the top 30% of syllables per animal selected by pooled late pre-lesion and post-lesion variability, which allowed us to capture syllables that became unstable after lesion without selecting exclusively on post-lesion variability. This analysis retained 146 animal $\times$ syllable pairs across 21 birds, including 30 sham syllables, 52 lateral-only lesion syllables, and 64 medial+lateral lesion syllables.

Across selected syllables, many medial+lateral lesion syllables shifted above the late pre-lesion versus post-lesion identity line, indicating increased post-lesion phrase duration SD. This pattern was less apparent in sham and lateral-only lesion birds. At the animal level, sham birds showed little change in phrase duration variability, with median SD changing from 0.65 s in late pre-lesion song to 0.72 s post-lesion (median $\Delta = 0.009$ s; paired t-test, $t = 0.27$, $p = 0.402$, $n = 4$). Lateral-only lesion birds also did not show a significant increase, with median SD changing from 0.78 s to 0.69 s (median $\Delta = -0.077$ s; $t = -0.94$, $p = 0.811$, $n = 8$). In contrast, birds with complete or partial medial+lateral lesions showed a significant post-lesion increase in phrase duration variability, with median SD increasing from 0.69 s to 1.03 s (median $\Delta = 0.109$ s; $t = 2.04$, $p = 0.0376$, $n = 9$).

This increase was also visible across time. Sham and lateral-only lesion groups showed relatively stable phrase duration SD across the recording window, whereas the medial+lateral lesion group showed a sustained post-lesion elevation and a pronounced late post-lesion increase. Together, these results indicate that lesions involving medial AFP circuitry selectively destabilized phrase timing in adult canary song.

The increase in phrase-duration standard deviation was robust to the high-variability threshold: pooled medial+lateral lesion birds showed significant post-lesion increases when selecting the top 20%, 30%, or 40% most variable syllables (Supplemental Table X, Supplemental Fig.X). In an independent validation that selected syllables using pre-lesion variability alone, pooled medial+lateral birds showed the same direction of effect, whereas sham and lateral-only birds did not increase, although this conservative animal-level validation did not reach $p < 0.05$.

We next asked whether the magnitude of the phrase-duration variability change scaled with lesion extent. Across animals, effective Area X lesion percentage was not significantly correlated with the median change in phrase-duration standard deviation across the selected top 30% high-variability syllables (Spearman $\rho = 0.36$, $p = 0.10$, $n = 21$ animals; Supplemental Fig. X). Thus, although birds with medial+lateral AFP lesions showed increased phrase-duration variability, lesion volume alone did not significantly predict the magnitude of this effect.

To determine whether increased phrase duration variability was transient or sustained, we examined phrase duration standard deviation across days relative to lesion (Fig. 3E). Daily variability was averaged using a 7-day rolling median, with the shaded region indicating the rolling interquartile range. Both sham controls and lateral AFP only lesions remained relatively stable across the recording period. In contrast, birds with medial and lateral AFP lesions showed elevated phrase duration variability across the post-lesion period. This suggests that the increase in phrase duration variability was sustained over the duration of the study rather than restricted to the immediate aftermath of lesion surgery.

Lesions to the medial AFP produced subtle, but detectable, changes to syllable acoustic structure

Higher overlap of a syllable's timebins in TweetyBERT's latent space suggests higher similarity in acoustic structure; likewise, reduced overlap suggests a shift in overall acoustic structure or greater acoustic variability (Vengrovski et al., 2026). To test whether AFP lesions altered the acoustic structure of syllables, we quantified acoustic latent-space consistency using the Bhattacharyya coefficient (BC). For statistical analysis, cluster-level BCs were summarized as bird-level medians, so each bird contributed one pre-treatment and one post-treatment value per comparison.

Birds with combined medial and lateral AFP lesions had a small, but significant post-lesion reduction in BC ($n = 8$ birds; median pre = 0.9910, median post = 0.9740, median $\Delta BC = -0.0100$; $p = 0.0391$). This reduced overlap indicates that medial and lateral AFP lesions were associated with lower consistency in the acoustic latent space. It remains to be determined if this reduced consistency is the result of an increase in trial-to-trial syllable variability, or a drift in syllable form post-lesion. Sham saline injection birds showed no significant pre/post change in BC ($n = 4$ birds; median pre = 0.9870, median post = 0.9869, median $\Delta BC = +0.0012$; paired Wilcoxon signed-rank test, $p = 0.875$; Fig. 4C). Birds with lateral AFP lesions alone also did not show a significant pre/post change, although the median post-treatment value was lower than the pre-treatment value ($n = 5$ birds; median pre = 0.9839, median post = 0.9732, median $\Delta BC = -0.0086$; $p = 0.625$).

Next, we investigated whether this post-lesion acoustic change was concentrated in syllable clusters that exhibited the strongest lesion-associated variation in repeat time. Within medial and lateral lesion birds, the top 30% highest phrase duration variability clusters showed a significant reduction in BC after lesion (median pre = 0.9931, median post = 0.9761, median $\Delta BC = -0.0121$; paired Wilcoxon signed-rank test, $p = 0.0234$; Fig. 4D). In contrast, the remaining non-top-variability clusters did not show a significant pre/post change (median pre = 0.9921, median post = 0.9873, median $\Delta BC = -0.0047$; $p = 0.156$). These results suggest that the reduction in acoustic consistency post-lesion was most pronounced in syllables that exhibited a stuttering-like phenotype.

Discussion

Stuttering-like increase in phrase duration variability

When the medial and lateral portion of the Anterior Forebrain Pathway (AFP) was lesioned, adult canaries developed a post-lesion stuttering-like behavior. That is, the variability in the time spent continuously repeating a syllable before transitioning to the next syllable increased. This stuttering-like effect was absent in sham control birds, as well as birds whose lesions were confined to the lateral AFP. These results suggest that the medial thalamocortico-basal ganglia pathway contributes to the timing of phrase transitions in canaries.

The increase in phrase duration variability increased throughout the post-lesion recording period rather than resolving shortly after surgery. It was accompanied by subtly reduced acoustic consistency in latent space. Together, these results demonstrate that the medial AFP continues to regulate syllable-level timing in adult canary song, challenging the view that the AFP is dispensable once song is learned.

The progression of the stuttering effect we observed differed from similar stuttering findings from songbirds of different species. Lesions in Bengalese finches produced a stuttering effect that peaked shortly after surgery and then disappeared (Kobayashi et al., 2001). Lesions in zebra finches appeared to worsen over time, with the magnitude of stuttering correlated with higher post-lesion recovery in the basal ganglia nucleus lateral Area X (Kubikova et al., 2014). In this study, we observed that a majority of the impacted phrase durations remained highly variable for the remainder of the study (Fig. 3E). The prolonged duration suggests that this effect was not due to changes in plasticity induced by neurons firing, then dying in the excitotoxic lesions in downstream song nuclei (e.g., RA).

Interestingly, effective Area X lesion extent was not significantly correlated with the animal-level median change in phrase-duration standard deviation across selected high-variability syllables (Supplemental Fig. X). This result corresponds to the findings of Kobayashi and colleagues, who found that lesion size did not correlate with the degree of stuttering-like effect in Bengalese finches (Kobayashi et al., 2001). Instead, our results suggest that lesion hit type, particularly involvement of medial AFP circuitry, was more informative than lesion volume alone. Excitotoxic lesions only made to brain regions in the Lateral AFP did not produce an increase in outlier syllable repeat time compared to sham saline lesions. In contrast, lesions with hits to both the lateral and medial portions of the AFP resulted in a significant increase in the tails of the syllable repeat time distribution. These results correspond with early Area X lesions performed in adult canaries. The study by Kobayashi and colleagues did not distinguish between lateral and medial Area X hits, as this work preceded the publication of neuroanatomical findings by Kubikova and colleagues, who used ZENK expression to suggest the existence of the putative medial AFP (Kubikova et al., 2007). Our findings may suggest that their stuttering effect was due to damage to the medial portions of Area X or the AFP. In zebra finches, lesions to lateral Area X produced a stuttering-like behavior in a subset of adult zebra finches: only zebra finches whose pre-lesion song ended with repeat syllables produced songs with stuttering post-lesion (Kubikova et al., 2014). Further, only excitotoxic lesions produced this stuttering effect; electrolytic lesions to lateral Area X did not show any significant impact on syllable repeats. The authors also found that as the birds recovered a greater percentage of their lesioned lateral Area X, their stuttering effect increased. Across all lesioned birds, the authors found a reduction in syllable duration and

an increase in stereotypy of syllable sequence (Kubikova et al., 2014). Nottebohm and colleagues described finding no impact on song after unilateral lesions to Area X. Interestingly, their histological figures indicate that they lesioned the lateral portion of Area X and did not lesion medial Area X (Nottebohm et al., 1976), suggesting alignment with our study's findings in the canary song system.

Comparison of songbird stuttering-like behavior to human stuttering

The stuttering-like behavior we observed shares several features with human stuttering. In both songbirds and humans, stuttering can be framed as difficulty transitioning out of a repeated motor pattern, leading to repetitions or prolongations of syllables. Adults who stutter exhibit increased variability in rhythmic timing, resulting in abnormal speech and nonspeech movement kinematics (Daliri et al., 2014; Max et al., 2003; Max & Yudman, 2003). Dopaminergic signaling within basal ganglia circuits is implicated in human stuttering, which is consistent with our targeting of Area X, a songbird basal ganglia nucleus containing Medium Spiny Neurons. The increase in phrase duration variability we observed corresponds with evidence that stuttering is associated with heightened motor and timing variability, including more variable sensorimotor integration and more unstable speech movements (Olander et al., 2010; Sares et al., 2018; Wiltshire et al., 2021). As with all animal models of human diseases, there are limitations and differences. Human stuttering typically begins in early childhood and often remits (Cornez et al., 2020; Franken et al., 2018; Reilly et al., 2013), whereas our effects were observed in adult canaries singing crystallized song and persisted across the post-lesion recording period. Human stuttering is also most commonly concentrated at word-initial or utterance-initial positions (Natke et al., 2004), whereas the phrase duration increases we observed were distributed across syllables in the repertoire and were not preferentially concentrated at the first syllable of a song. The convergence on basal ganglia-dependent timing/sequencing dysfunction makes the medial AFP an attractive model for dissecting stuttering-like motor sequencing deficits (Kubikova et al., 2014; Moorman et al., 2021).

Detectable changes to acoustic variability

These results fit the broader view that the AFP is not simply required for producing crystallized adult song but contributes to variability, plasticity, and sequence regulation. Prior work showed that LMAN lesions can stabilize adult song after deafening, suggesting that AFP output contributes to ongoing song variability and experience-dependent modification rather than basic song execution (Brainard & Doupe, 2001). Similarly, LMAN lesions in adult Bengalese finches reduce syllable-structure variability, supporting a role for AFP output in regulating acoustic variability (Hampton et al., 2009). We found that sham and lateral-only lesion birds showed little change, whereas medial and lateral AFP lesions resulted in a small, but significant reduction in post-lesion acoustic latent space overlap. This is consistent with the idea that stronger AFP disruption may destabilize adult song structure.

The results also align with Area X lesion studies showing effects on adult song tempo, syllable sequencing, and repetition. Prior work in zebra finches found that adult Area X damage altered tempo and global syllable sequencing and increased phrase duration in birds predisposed to

repeat syllables (Kubikova et al., 2014). Our results extend this by showing that the phrase types most associated with increased duration variability also show subtly reduced acoustic consistency, suggesting that stuttering-like repetition is accompanied by acoustic latent-space destabilization rather than only a transition-timing deficit.

The stuttering-like effect produced by the lesions appears selective rather than global. Recent MMAN lesion work found increased syntax variability with syllable acoustic structure largely intact (Hampton et al., 2009; Koparkar et al., 2024) found that LMAN lesions differentially affected syllable-structure versus sequence variability. Our results showed that syllables whose phrase duration variability increased post-lesion had a subtle change to acoustic structure. These findings support a model in which the AFP contributes to both the progression of repeated syllables and their consistent production.

Conclusion and future directions

Our findings demonstrate that the medial AFP continues to play an active role in regulating the phrase structure of adult canary song. We performed bilateral excitotoxic lesions targeting both medial and lateral circuits in the AFP in breeding-season canaries and built a novel analysis method to process thousands of songs efficiently. We found that lesions involving the medial AFP produced a sustained stuttering-like behavior in a subset of syllables, characterized by increased variability in syllable repeat time. Additionally, we found that medial AFP lesions reduced the acoustic consistency of the stuttered syllables. Lesions to only the lateral AFP did not produce this effect, suggesting that the medial thalamocortico-basal ganglia circuit contributes to the ongoing maintenance of phrase-level timing in crystallized song, as well as syllable-level stereotypy. Rather than being a passive remnant of the learning process, the medial pathway appears to remain functionally engaged in performance, actively constraining the variability of repeated motor elements.

Moving forward, electrophysiological recordings from medial AFP nuclei and downstream Song Motor Pathway nuclei (HVC, RA) during singing will allow us to investigate the neural dynamics underlying the changes we observed. Characterizing how medial AFP activity is coordinated with HVC and RA during phrase transitions, and how this coordination breaks down following lesion, would provide mechanistic insight into how basal ganglia and cortical circuits jointly maintain the temporal structure of complex learned motor sequences. More broadly, the medial AFP offers a tractable and well-defined circuit for investigating the neural basis of motor sequencing disorders, such as stuttering, in a system where causal intervention, high-resolution behavioral quantification, and single-unit physiology can be combined.

Code Availability

Analysis code is available at https://github.com/hulseylvincentr/AFP_lesion_paper. TweetyBERT is available at https://github.com/georgeevenven/tweety_bert. The TweetyNET song detector is available at https://github.com/georgeevenven/tweety_net_song_detector.

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Author contributions

T.J.G. and M.R.H.-V. conceived of the study, designed the experiment, and co-wrote the manuscript. M.R.H.-V. performed the surgeries, collected data, cared for the birds, and inspected and analyzed the data. T.J.G. advised data analysis and interpretation. G.V. built TweetyBERT and the decoder used for analysis, gave feedback on analysis, and revised the paper. E.S. cared for birds, developed the Nissl staining protocol, and assisted in building the song detection algorithm with G.V., and revised the paper.

Declaration of interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used ChatGPT and Claude solely to rephrase, simplify, and improve the manuscript's prose and grammar. All suggestions were reviewed and approved by the authors before submission.

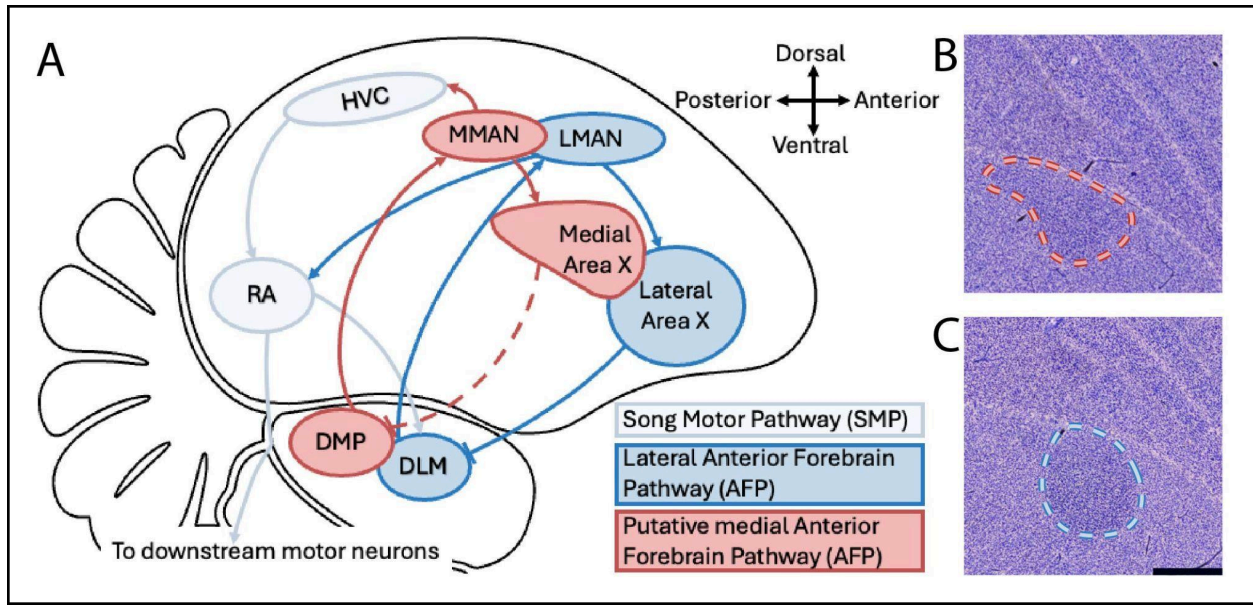


Figure 1. Anatomical reference for the medial and lateral Anterior Forebrain Pathway (AFP) targeting. A) Sagittal map of the organization of the AFP and the Song Motor Pathway. The lateral AFP is shown in blue, and the putative medial AFP is shown in red. B) Representative sagittal Nissl-stained section showing intact medial Area X outlined by a red dashed line. C) Representative sagittal Nissl-stained section of lateral Area X, outlined by a blue dashed line. The scale bar indicates 500 micrometers.

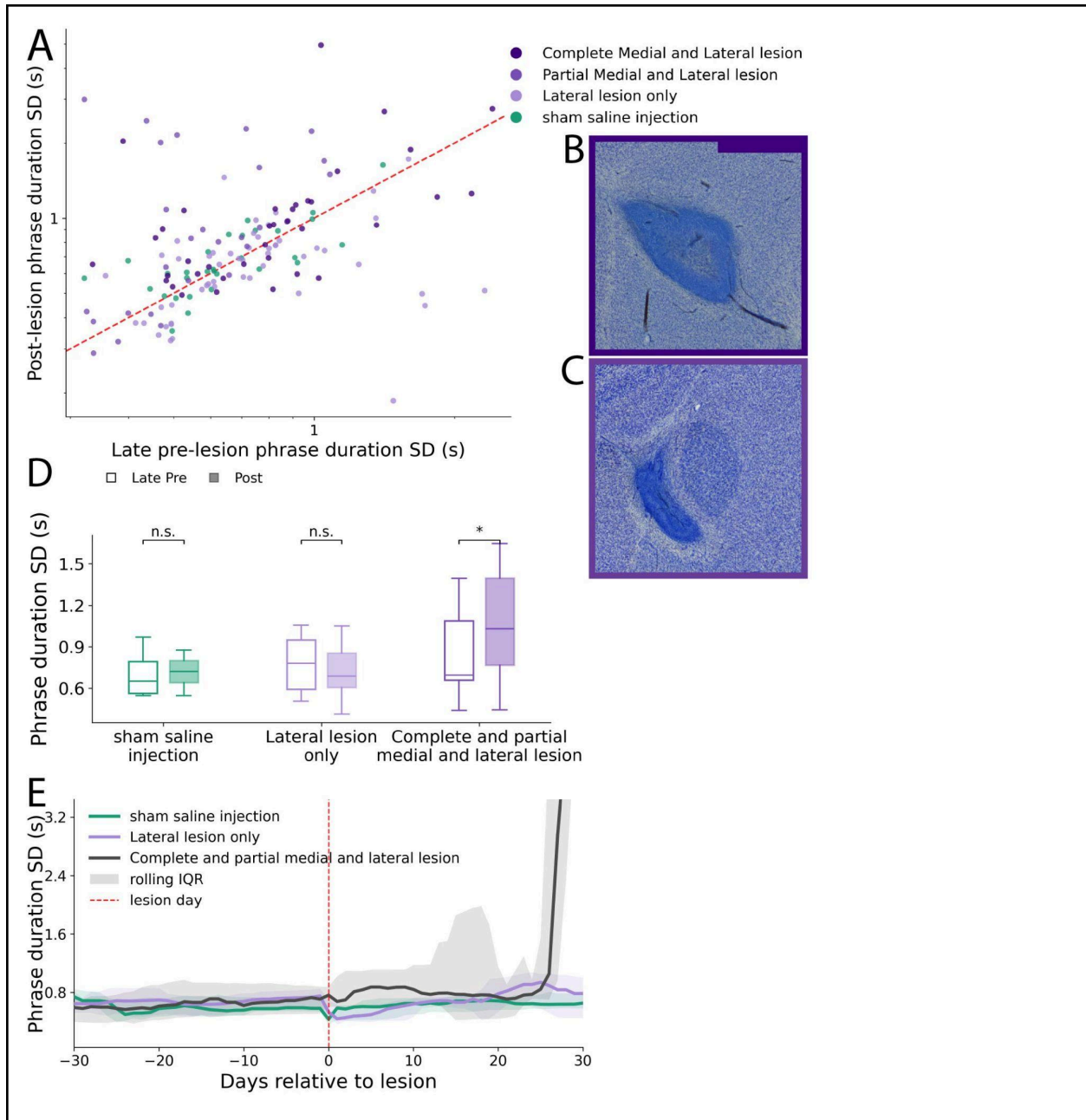


Figure 3. Lesions involving medial AFP circuitry increase phrase duration variability.

A) Late pre-lesion versus post-lesion phrase duration standard deviation for selected high-variability syllables. Syllables were selected within each animal as the top 30% ranked by pooled late pre-lesion and post-lesion variability. Each point represents one animal $\times$ syllable pair. Points above the dashed identity line indicate greater post-lesion phrase duration variability. Points are color-coded by lesion hit type.

B) Representative Nissl-stained section from a bird with a complete medial+lateral lesion.

C) Representative Nissl-stained section from a bird with a partial medial+lateral lesion.

D) Animal-level phrase duration SD for selected syllables in late pre-lesion and post-lesion song. Selected syllables were averaged within each bird before statistical testing. Sham and lateral-only lesion birds did not show significant increases in phrase duration SD, whereas birds with complete or partial medial+lateral lesions showed a significant post-lesion increase.

Boxes show the distribution across animals; points represent individual animals. One-tailed paired t-tests were performed at the animal level.

E) Time course of phrase duration SD across days relative to lesion. Daily phrase duration variance values were converted to SD in seconds and filtered to the same selected animal $\times$ syllable pairs used in panels A and D. Lines show the 7-day rolling median for each lesion group, shaded regions show the rolling interquartile range, and the red dashed line marks lesion day. Sham and lateral-only lesion groups remained relatively stable, whereas the medial+lateral lesion group showed sustained post-lesion elevation in phrase duration variability.

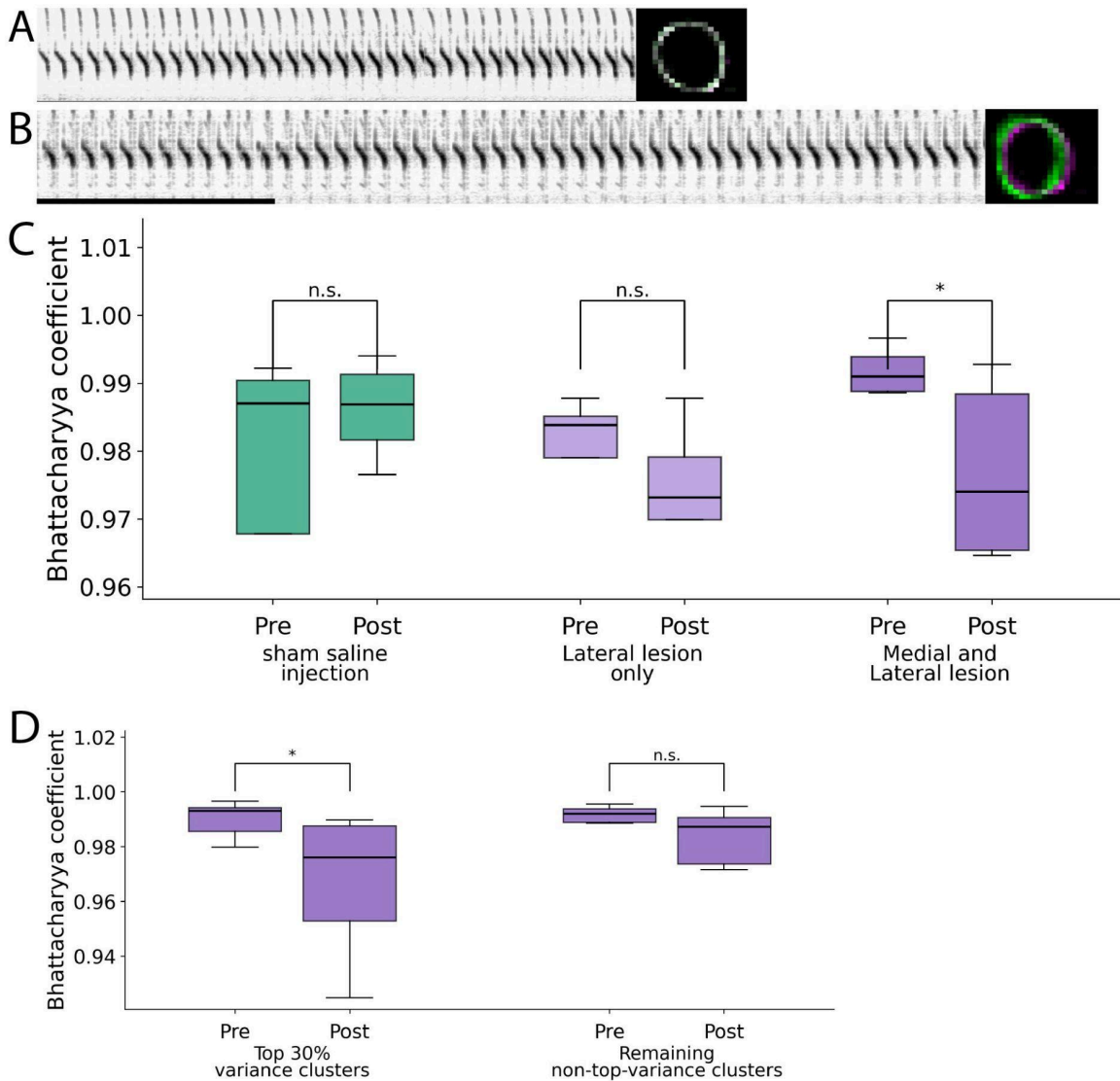


Figure 4. Medial and lateral AFP lesions cause modest but detectable acoustic changes in stuttered syllables.

A) Representative pre-lesion phrase spectrogram and corresponding UMAP density overlap for one syllable cluster. Early and late pre-lesion time bins were evenly sampled; white

indicates high overlap. Pre-lesion BC = 0.995. B) Representative late post-lesion spectrograms from the same syllable show reduced stereotypy; scale bar, 1 s. Corresponding early versus late post-lesion UMAP overlap was reduced (BC = 0.623). C) Within-period BC comparison by lesion hit type. Sham ($p = 0.875$) and lateral-only lesion birds ($p = 0.625$) showed no significant change, whereas medial and lateral AFP lesion birds showed a significant post-lesion decrease in BC ($p = 0.039$). Lower BC indicates reduced latent-space overlap. D) In medial and lateral AFP lesion birds, stuttered syllables showed a significant BC decrease ($p = 0.023$), whereas non-stuttered syllables did not ($p = 0.156$).

References

- Bhattacharyya, A. (1943). On a measure of divergence between two statistical populations defined by their probability distributions. *Bulletin of the Calcutta Mathematical Society*, 35, 99–109.
- Bottjer, S. W., Miesner, E. A., & Arnold, A. P. (1984). Forebrain lesions disrupt development but not maintenance of song in passerine birds. *Science*, 224(4651), 901–903.
- Brainard, M. S., & Doupe, A. J. (2001). Postlearning consolidation of birdsong: stabilizing effects of age and anterior forebrain lesions. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 21(7), 2501–2517.
- Cornez, G., Collignon, C., Müller, W., Ball, G. F., Cornil, C. A., & Balthazart, J. (2020). Seasonal changes of perineuronal nets and song learning in adult canaries (*Serinus canaria*). *Behavioural Brain Research*, 380, 112437.
- Craig-McQuaide, A., Akram, H., Zrinzo, L., & Tripoliti, E. (2014). A review of brain circuitries involved in stuttering. *Frontiers in Human Neuroscience*, 8, 884.
- Daliri, A., Prokopenko, R. A., Flanagan, J. R., & Max, L. (2014). Control and prediction components of movement planning in stuttering versus nonstuttering adults. *Journal of Speech, Language, and Hearing Research*, 57(6), 2131–2141.
- Fee, M. S., & Goldberg, J. H. (2011). A hypothesis for basal ganglia-dependent reinforcement learning in the songbird. *Neuroscience*, 198, 152–170.
- Franken, M.-C. J. P., Koenraads, S. P. C., Holtmaat, C. E. M., & van der Schroeff, M. P. (2018). Recovery from stuttering in preschool-age children: 9 year outcomes in a clinical

- population. *Journal of Fluency Disorders*, 58, 35–46.
- Goldberg, J. H., & Fee, M. S. (2011). Vocal babbling in songbirds requires the basal ganglia-recipient motor thalamus but not the basal ganglia. *Journal of Neurophysiology*, 105(6), 2729–2739.
- Hampton, C. M., Sakata, J. T., & Brainard, M. S. (2009). An avian basal ganglia-forebrain circuit contributes differentially to syllable versus sequence variability of adult Bengalese finch song. *Journal of Neurophysiology*, 101(6), 3235–3245.
- Hara, E., Kubikova, L., Hessler, N. A., & Jarvis, E. D. (2007). Role of the midbrain dopaminergic system in modulation of vocal brain activation by social context. *The European Journal of Neuroscience*, 25(11), 3406–3416.
- Huang, J. (n.d.). *Thionin staining for frozen tissues*.
<https://www.kumc.edu/documents/kiddrc/Thionin-Staining-frozen-tissues.pdf>
- Jarvis, E. D. (2019). Evolution of vocal learning and spoken language. *Science*, 366(6461), 50–54.
- Kobayashi, K., Uno, H., & Okanoya, K. (2001). Partial lesions in the anterior forebrain pathway affect song production in adult Bengalese finches. *Neuroreport*, 12(2), 353–358.
- Koparkar, A., Warren, T. L., Charlesworth, J. D., Shin, S., Brainard, M. S., & Veit, L. (2024). Lesions in a songbird vocal circuit increase variability in song syntax. *eLife*, 13.
<https://doi.org/10.7554/eLife.93272>
- Kubikova, L., Bosikova, E., Cvikova, M., Lukacova, K., Scharff, C., & Jarvis, E. D. (2014). Basal ganglia function, stuttering, sequencing, and repair in adult songbirds. *Scientific Reports*, 4, 6590.
- Kubikova, L., Turner, E. A., & Jarvis, E. D. (2007). The pallial basal ganglia pathway modulates the behaviorally driven gene expression of the motor pathway. *The European Journal of Neuroscience*, 25(7), 2145–2160.
- Lashley, K. S. (1951). *The problem of serial order in behavior* (Vol. 21). Bobbs-Merrill Oxford.

- Markowitz, J. E., Ivie, E., Kligler, L., & Gardner, T. J. (2013). Long-range order in canary song. *PLoS Computational Biology*, 9(5), e1003052.
- Max, L., Caruso, A. J., & Gracco, V. L. (2003). Kinematic analyses of speech, orofacial nonspeech, and finger movements in stuttering and nonstuttering adults. *Journal of Speech, Language, and Hearing Research*, 46(1), 215–232.
- Max, L., & Yudman, E. A. (2003). Accuracy and variability of isochronous rhythmic timing across motor systems in stuttering versus nonstuttering individuals. *Journal of Speech, Language, and Hearing Research*, 46(1), 146–163.
- Moorman, S., Ahn, J.-R., & Kao, M. H. (2021). Plasticity of stereotyped birdsong driven by chronic manipulation of cortical-basal ganglia activity. *Current Biology: CB*, 31(12), 2619–2632.e4.
- Natke, U., Sandrieser, P., van Ark, M., Pietrowsky, R., & Kalveram, K. T. (2004). Linguistic stress, within-word position, and grammatical class in relation to early childhood stuttering. *Journal of Fluency Disorders*, 29(2), 109–122.
- Nottebohm, F., Stokes, T. M., & Leonard, C. M. (1976). Central control of song in the canary, *Serinus canarius*. *The Journal of Comparative Neurology*, 165(4), 457–486.
- Olander, L., Smith, A., & Zelaznik, H. N. (2010). Evidence that a motor timing deficit is a factor in the development of stuttering. *Journal of Speech, Language, and Hearing Research*, 53(4), 876–886.
- Reilly, S., Onslow, M., Packman, A., Cini, E., Conway, L., Ukoumunne, O. C., Bavin, E. L., Prior, M., Eadie, P., Block, S., & Wake, M. (2013). Natural history of stuttering to 4 years of age: a prospective community-based study. *Pediatrics*, 132(3), 460–467.
- Reiner, A., Perkel, D. J., Bruce, L. L., Butler, A. B., Csillag, A., Kuenzel, W., Medina, L., Paxinos, G., Shimizu, T., Striedter, G., Wild, M., Ball, G. F., Durand, S., Güntürkün, O., Lee, D. W., Mello, C. V., Powers, A., White, S. A., Hough, G., ... Avian Brain Nomenclature Forum. (2004). Revised nomenclature for avian telencephalon and some related brainstem nuclei.

The Journal of Comparative Neurology, 473(3), 377–414.

Sares, A. G., Deroche, M. L. D., Shiller, D. M., & Gracco, V. L. (2018). Timing variability of sensorimotor integration during vocalization in individuals who stutter. *Scientific Reports*, 8(1), 16340.

Sohrabji, F., Nordeen, E. J., & Nordeen, K. W. (1990). Selective impairment of song learning following lesions of a forebrain nucleus in the juvenile zebra finch. *Behavioral and Neural Biology*, 53(1), 51–63.

Tchernichovski, O., Nottebohm, F., Ho, C. E., Pesaran, B., & Mitra, P. P. (2000). A procedure for an automated measurement of song similarity. *Animal Behaviour*, 59(6), 1167–1176.

Vengrovski, G., Hulse-Vincent, M. R., Bemrose, M. A., & Gardner, T. J. (2026). TweetyBERT: Automated parsing of birdsong through self-supervised machine learning. *Patterns*, 0(0), 101491.

Wiltshire, C. E. E., Chiew, M., Chesters, J., Healy, M. P., & Watkins, K. E. (2021). Speech movement variability in people who stutter: A vocal tract magnetic resonance imaging study. *Journal of Speech, Language, and Hearing Research*, 64(7), 2438–2452.